

Pumping lemma (For Regular languages)

(39)

- Pumping lemma is used to prove that a language is NOT Regular
- It cannot be used to prove that a language is Regular.

If A is a Regular Language, then A has a Pumping Length ' p ' such that any string ' s ' where $|s| \geq p$ may be divided into 3 parts, $s = xyz$ such that the following conditions must be true :

$$(1) xy^iz \in A \text{ for every } i \geq 0$$

$$(2) |y| > 0$$

$$(3) |xy| \leq p$$

* To prove that a language is not regular using Pumping Lemma, follow the below steps : (Prove using contradiction)

- Assume that A is regular
- It has to have a pumping length (say p)
- All strings longer than p can be pumped $|s| \geq p$
- Now find a string ' s ' in A such that $|s| \geq p$
- Divide s into $x y z$
- show that $xy^iz \notin A$ for some i
- Then consider all ways that s can be divided into $x y z$
- show that none of these can satisfy all the 3 pumping conditions at the same time
- s cannot be pumped == CONTRADICTION